

4. (a) Compounds are formed when atoms of different elements combine.

Draw **one** line from each compound to its formula.

[2]

**Compound**

**Formula**

carbon dioxide

CO

CO<sub>2</sub>

Ca<sub>2</sub>O

sodium hydroxide

NaOH

NaNO<sub>3</sub>

Na<sub>2</sub>CO<sub>3</sub>

- (b) Magnesium oxide contains the ions Mg<sup>2+</sup> and O<sup>2-</sup>.

Give the formula of magnesium oxide.

[1]

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- (c) Calcium sulfate has the formula  $\text{CaSO}_4$ .

Calculate the relative formula mass ( $M_r$ ) of calcium sulfate.

[2]

$$A_r(\text{Ca}) = 40$$

$$A_r(\text{S}) = 32$$

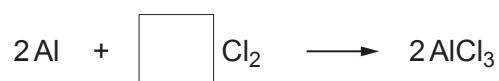
$$A_r(\text{O}) = 16$$

$$M_r = \dots\dots\dots$$

- (d) Aluminium chloride,  $\text{AlCl}_3$ , is formed by the reaction of aluminium with chlorine.

Write a number in the box to balance the equation for this reaction.

[1]

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